The most accurate model was looking at the wrong thing

PneumoScan · project summary · live demo pneumoscan-nine.vercel.app · code github.com/Prathamesh250205/PneumoScan-CNN

Three ImageNet-pretrained CNNs were fine-tuned to detect pneumonia in pediatric chest X-rays (Kermany et al.), each with three training seeds. ResNet18 was the most accurate (88%), yet only 47% of its Grad-CAM attention fell on the lungs, barely above the 39% expected by chance. MobileNetV2 put 67% there, in every seed. All three rank cases equally well (AUC 0.954–0.960). On 656 adult X-rays from another hospital every model degraded, and MobileNetV2 degraded least.

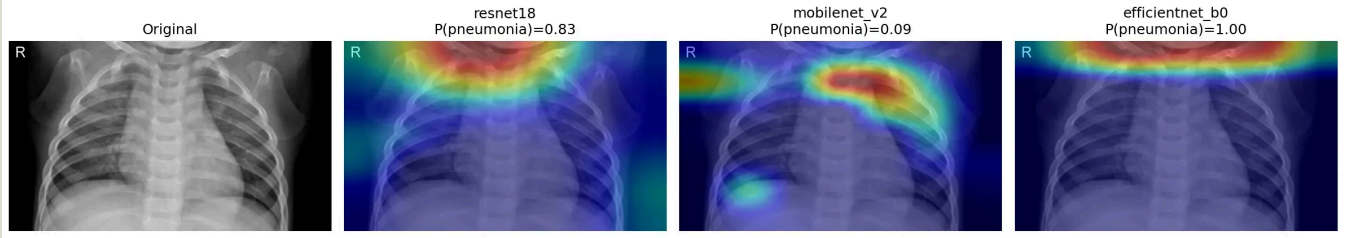


Fig. 1 A normal X-ray. ResNet18 predicts pneumonia (P = 0.83) with attention on the neck and upper mediastinum. MobileNetV2 correctly predicts normal (P = 0.09) with attention on the lung.

1 · In-distribution performance (mean ± SD over 3 seeds)

Model	Accuracy	F1	AUC	Sens.	Spec.
ResNet18	**88.2 ± 2.9%**	0.913 ± 0.019	0.954 ± 0.013	0.989	0.704 ± 0.083
MobileNetV2	83.9 ± 2.6%	0.886 ± 0.016	0.956 ± 0.025	0.996	0.578 ± 0.072
EfficientNet-B0	84.3 ± 0.6%	0.888 ± 0.004	0.960 ± 0.002	0.996	0.588 ± 0.017

Kermany test split, n = 624 (234 normal, 390 pneumonia), threshold 0.5. The original single run gave ResNet18 0.978 AUC; retraining gives 0.940–0.966, so that lead was a lucky draw. For the served ResNet18, a threshold chosen for 95% sensitivity by cross-validation (t = 0.946, used by the live demo) raises specificity from 0.68 to 0.88.

2 · Where the models look

Lungs, clavicles and scapulae were segmented with torchxrayvision's ChestX-Det PSPNet. For each image we measured the share of Grad-CAM energy (last conv block, predicted class) inside the lung mask. Normal X-rays are the clean comparison, because the adult-trained segmenter drops some consolidated lung on pediatric pneumonia X-rays.

Normal X-rays · mean ± SD (seeds)	Pediatric, n = 234	Adult NIH, n = 150
chance (lung share of image)	*38.5%*	*33.4%*
ResNet18	46.8 ± 5.5% (47.8 · 41.0 · 51.8)	39.0 ± 5.3% (38.2 · 34.2 · 44.7)
MobileNetV2	**67.4 ± 3.7%** (68.6 · 63.3 · 70.3)	**58.3 ± 2.3%** (55.7 · 60.1 · 59.1)
EfficientNet-B0	46.9 ± 7.1% (55.0 · 41.8 · 43.9)	26.8 ± 15.0% (44.1 · 16.7 · 19.7)

Every MobileNetV2 seed beats every ResNet18 seed on both datasets (pediatric gap +20.8, +22.3, +18.5 points). Correction to this project's first, visual claim: the served ResNet18's attention on clavicles and scapulae (4.7%) is below their 6.4% share of the image. It isn't on the shoulders; it's just not on the lungs.

3 · External test: another hospital, adult patients

656 X-rays from NIH ChestX-ray14 (batch 7) with radiologist-adjudicated labels (Majkowska et al., 2020): 364 airspace opacity, 292 with all expert labels negative, mostly AP views.

Model (mean ± SD, 3 seeds)	Pediatric AUC	NIH AUC	Drop
ResNet18	0.954 ± 0.013	0.794 ± 0.023	−0.160
MobileNetV2	0.956 ± 0.025	**0.822 ± 0.024**	**−0.134**
EfficientNet-B0	0.960 ± 0.002	0.758 ± 0.012	−0.202

- The lung-focused model degrades least on average. Seed by seed, MobileNetV2 − ResNet18 NIH AUC is −0.017, +0.078 and +0.026: a trend, not a significant result, with three seeds.
- Thresholds don't transfer. At the pediatric 0.946, the served ResNet18 flags 80% of healthy adults (specificity 0.199).
- The shift mixes a new hospital with a new age group, and it hurts every model.

Method and system

- **Training:** PyTorch, ImageNet weights, 2-class head, Adam 1e-4, 10 epochs, flips and ±10° rotation; 3 seeds per model (Kaggle T4).
- **Preprocessing:** grayscale → 3 channels → 224^2 bilinear → ImageNet normalisation. Identical in training and serving.
- **Statistics:** stratified bootstrap (2,000 draws), paired across models on the same images.
- **Serving:** ONNX Runtime + FastAPI as a Vercel function. The ONNX graphs also output the class activation map, which equals Grad-CAM for global-average-pool heads (verified correlation 1.0), so live heatmaps need no backward pass.
- **Testing:** Playwright end-to-end test on desktop and mobile, run against local and production.

Limitations

- Three seeds per model: enough to show stability, too few for formal tests between architectures.
- The external test mixes a hospital shift with an age shift (children → adults).
- Lung masks come from a segmentation model, not radiologists.
- The validation split has 16 images, so operating points were cross-validated on the test set and are optimistic for new sites.

> **Research demonstration only. Not a medical device.** Not validated for clinical use and not cleared by any regulator. Do not use for diagnostic decisions.

Data: Kermany et al., *Cell* 2018 · Wang et al., CVPR 2017 (ChestX-ray14) · Majkowska et al., *Radiology* 2020 · Cohen et al., MIDL 2022 (TorchXRayVision).